

# A mutation-sensitive, multiplexed and amplification-free detection of nucleic acids by stretching single-molecule tandem hairpin probes

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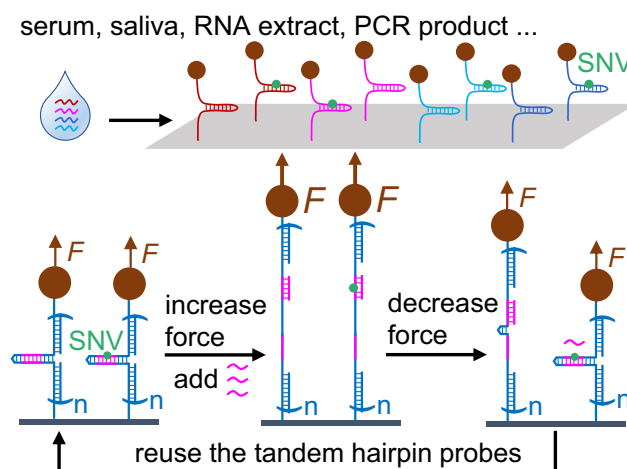
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## ABSTRACT

The detection of nucleic acid sequences in parallel with the discrimination of single nucleotide variations (SNVs) is critical for research and clinical applications. A few limitations make the detection technically challenging, such as too small variation in probe-hybridization energy caused by SNVs, the non-specific amplification of false nucleic acid fragments and the few options of dyes limited by spectral overlaps. To circumvent these limitations, we developed a single-molecule nucleic acid detection assay without amplification or fluorescence termed THREF (hybridization-induced tandem DNA hairpin refolding) based on multiplexed magnetic tweezers. THREF can detect DNA and RNA sequences at femtomolar concentrations within 30 min, monitor multiple probes in parallel, quantify the expression level of miR-122 in patient tissues, discriminate SNVs including the hard-to-detect G–U or T–G wobble mutations and reuse the probes to save the cost. In our demonstrative detections using mock clinic samples, we profiled the let-7 family microRNAs in serum and genotyped SARS-CoV-2 strains in saliva. Overall, the THREF assay can discriminate SNVs with the advantages of high sensitivity, ultra-specificity, multiplexing, reusability, sample hands-free and robustness.

## GRAPHICAL ABSTRACT



## INTRODUCTION

Nucleic acid-based biomarkers such as microRNAs (miRNAs) or pathogen-derived nucleic acid fragments in biological fluids are widely used targets for diagnostics because RNA and DNA enable highly specific detections by pairing with the complementary nucleotides. The nucleic acid test (NAT) is used in disease diagnoses such as infectious diseases, cancers, genetic disorders, and mitochondrial disorders (1,2). The fast and accurate detection of nucleic acids is extremely important for epidemiological surveillance, especially during the worldwide COVID-19 pandemic caused by SARS-CoV-2 (3).

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The current detection methods mainly rely on amplification and fluorescence techniques. Real-time quantitative polymerase chain reaction (RT-qPCR) is the most frequently adopted assay in clinics and laboratories due to its versatility, robustness, and especially its ultra-sensitivities. To get rid of the expensive thermocycler required for PCR, isothermal amplification-based methods such as loop-mediated isothermal amplification (LAMP) and recombinase polymerase amplification (RPA) were developed (4,5). However, methods incorporating amplification (RT-qPCR, LAMP and RPA) procedures require additional and professional operation steps that are time-consuming and prone to get false-positive results due to amplicon contaminations (6). Recently, CRISPR-based approaches combined with amplification were also developed with high sensitivity and convenience (7–9).

Single nucleotide variations (SNVs) of the human genome have wide-ranging effects on gene expression, RNA splicing, and protein recognition, and lead to far-reaching phenotypic changes, drug resistance, and cancer developments (10). Also, the SNVs of viral genes such as SARS-CoV-2 contribute to immune escape and reinfections (11). Thus, discrimination of SNVs is broadly required in the detection of nucleic acids. However, the small variations in probe-hybridization energy caused by SNVs especially the hard-to-detect G–U or T–G wobble base pairs make the identification of these mutations challenging, particularly in complex reaction environments (2). In comparison to single target detection, parallel discrimination of multiplex targets can offer more helpful information and promote understanding of the function of these targets in diseases (9,12).

Single-molecule manipulation platform is a newly developed strategy for the detection of proteins and nucleic acids (13–23). Compared with the bulk methods, technologies based on single-molecule manipulation, such as optical tweezers, acoustic tweezers and magnetic tweezers, use the applied tensile force or molecular extension of the sensing probe as the signal readout and thus avoid suffering from the need for quantification of fluorescence. While single-molecule detection technologies have offered the ultimate sensitivity at the precision of individual molecules theoretically, it requires quite a long time to detect the target at a very low concentration because diffusions limit the binding of the target to a single-molecule probe. Without amplification, the limit of detection (LOD) is only about 100 pM within half an hour of binding (14). To address this limitation, a recent study solved this sensitivity problem using DNA homopolymers that contained hundreds of binding sites for targets (15).

Here, we developed an assay termed THREF (hybridization-induced tandem DNA hairpin refolding failure) for the high-sensitive, SNV-discriminating, multiplex, amplification-free, fluorescence-free detection of nucleic acid sequences by combing multiplex magnetic tweezers (24–27) with reusable parallel unified tandem DNA hairpin probes. Briefly, the complete refolding of force-unfolded tandem hairpin probes could be effectively blocked upon hybridization of the sequence-matched targets (less effective for mutant targets), resulting in a relative probe length elongation defined as blocked length ( $L_B$ ) which is the signal readout of the assay. In the assay,

we first synthesized multiplex tandem hairpin probes with a unified degree of polymerization by synchronized rolling circle amplification (RCA) *in situ* (28,29). Afterward, we determined that the LOD of a single probe was femtomolar concentrations upon half an hour's binding, which would be further improved by monitoring multiple probes and by extending the binding time ( $t_B$ ). Taking advantage of our multiplexed magnetic tweezers, we manipulated multiple tandem hairpin probes in parallel. As the  $L_B$  is a function of the concentration of each target, we determined the expression level of a liver-specific miRNA has-mir-122-5p (miR-122) from hepatocellular carcinoma (HCC) foci and paracancerous tissues parallelly with a spike-in 20-nt RNA reference (RNA-Ref). By optimizing the probe-hybridization energy differences caused by a single-nucleotide change (substitution, deletion, or insertion) in the target, we rationally designed the sequence of probes and effectively discriminated the SNVs.

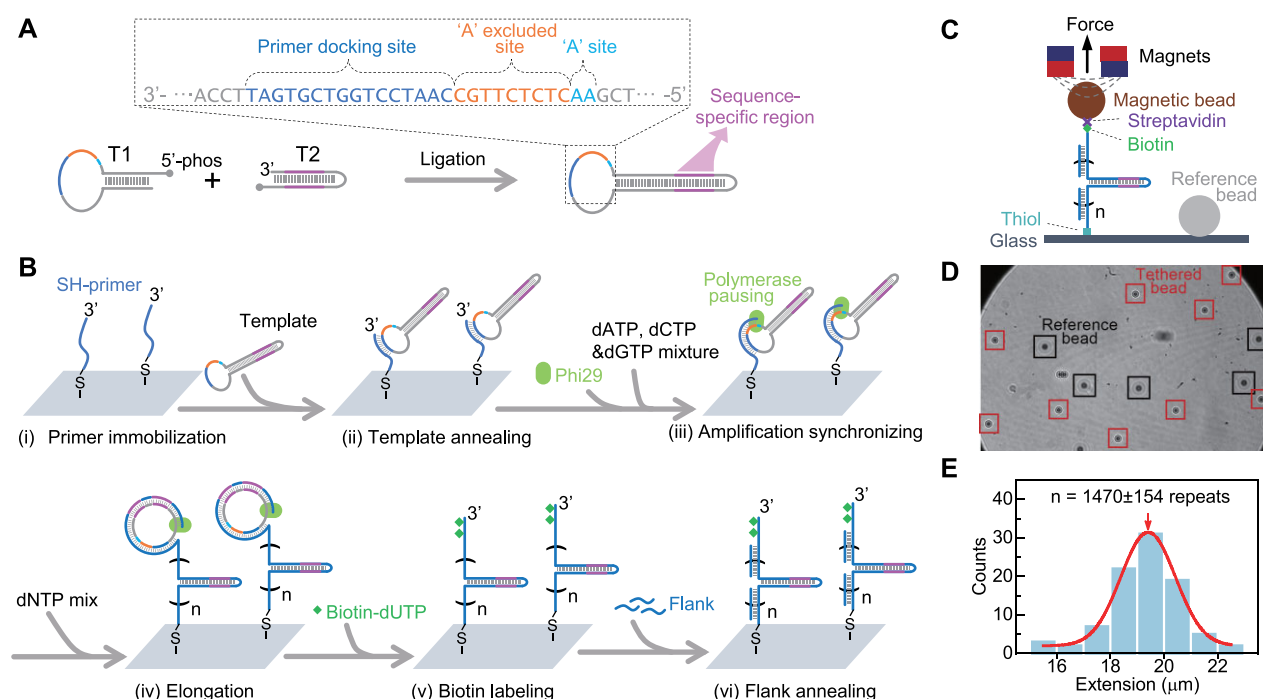
To demonstrate the utility and robustness of the THREF assay, we performed another two ultra-specific and multiplex detections. First, we simultaneously discriminated four homologous let-7 family miRNAs in fetal bovine serum (FBS). We obtained reliable results even with the interference of extremely high concentrations of mutant targets. Second, combining with an RNA fragmentation strategy, we genotyped three SARS-CoV-2 strains in human saliva in a single reaction. Thus, THREF is a powerful and multiplex molecular detecting method with significant specificities and potential applications ranging from biological studies to clinical applications.

## MATERIALS AND METHODS

### Magnetic tweezers and flow cell

Our magnetic tweezers were home-built based on an inverted microscope (IX73, Olympus) equipped with a 100 $\times$  oil immersion objective (PlanSAPO NA1.3, Olympus). The vertical position of the objective was controlled by a Piezo scanner (P-721, Physik Instrumente). A pair of separated NdFeB magnets was controlled by a stepper motor (L-509, Physik Instrumente) to apply constant forces to the superparamagnetic microbeads (M-270, Dynabeads) and stretch the DNA probes (Figure 1C). A high-brightness collimated LED-light source (MLS-201, Wuhan Yibo) was used to illuminate the flow cell through the gap between the magnets. The images were captured by a CCD camera (VC-12MC-M65EO, Vieworks) at 65 frames per second and analysed using home-written codes to obtain the 3D position of the microbeads. Note that, the magnetic moments of M-270 beads are quite uniform varying only <2%. Thus, the forces exerted on multiplexed probes in a view varied only <2%.

The flow cell was assembled by gluing two pre-treated coverslips of different sizes (the larger coverslip was 24  $\times$  72 mm and the smaller coverslip was 24  $\times$  50 mm) together using waterproof double-sided tapes (110  $\mu$ M thick, 3M), as described in our previous works (24,25,27). Briefly, before assembling, all coverslips were cleaned using a piranha solution. Once dried, the larger coverslips were functionalized with 1% APTES (Sigma) in methanol to coat the



**Figure 1.** Preparation of unified tandem DNA hairpin probes. (A) Preparation of circular ssDNA templates for RCA. Each type of circular template was prepared by ligating two stem-loop hairpin units T1 (shared by all types of templates) and T2. The loop of T1 contained three functional elements: a primer docking site (blue), the followed 'A' excluded site (orange) and 'A' site (cyan). T2 contained a sequence-specific region concealed in the stem (purple). (B) Flow chart to generate the tandem DNA hairpin probes by synchronized RCA in flow cells. (C) The probes were manipulated by home-built magnetic tweezers. One end of each probe was anchored to the SMCC-functionalized glass surface and the other end of the probe was attached to a streptavidin-functionalized superparamagnetic bead. A pair of NdFeB magnets was used to exert constant forces on the probe. Polystyrene beads were fixed to the surface as references to eliminate the drift. (D) A representative image was obtained with the multiplexed magnet tweezers. All magnetic beads (indicated by red squares) were tethered by single probes with a similar extension and thus had a similar diffraction pattern. The reference beads were indicated by black squares. (E) Distribution of probe extensions under a constant force of  $\sim 12$  pN. Fitting to normal distribution yielded that each probe had a polymerization number of  $1470 \pm 154$ .

glass surfaces with reactive amine groups. The APTES-functionalized flow cell can be sealed and stored at room temperature for more than half a year. Before immobilization of primers, we attached the polystyrene microbeads (3  $\mu\text{m}$  in diameter, Spherotech) to the APTES-functionalized surface of the flow cell by nonspecific adsorption in phosphate-buffered saline (PBS, Sangon Biotech). Here, the polystyrene microbeads were used as reference beads to eliminate the drift during the magnetic tweezing experiments (Figure 1C). After that, the APTES-functionalized surface was treated with Sulfo-SMCC (Thermo Fisher Scientific) in the flow cell, then 0.2 pM sulfhydryl labelled primer (SH-primer) was immobilized to the Sulfo-SMCC-functionalized surface in PBS. Finally, the internal surfaces of the flow cell were passivated by incubating with a blocking buffer that contained 1% bovine serum albumin (Sigma) and 1 mM 2-mercaptoethanol (Sigma) in PBS for 2 h. We resolved or diluted all analytes in 1 M NaCl, 10 mM Tris-HCl pH 7.5 and conducted the THREF experiments at 22  $^{\circ}\text{C}$ .

### Preparation of SARS-CoV-2 RNA fragments

The sequence of strain wild-type (NC\_045512.2) was downloaded from NCBI, and the sequences of strains B.1.617.2 (EPI\_ISL\_1470959) and BA.1 (EPI\_ISL\_13878991) were

downloaded from GISAID. Plasmids containing the full-length SARS-CoV-2 *spike* gene (3822 nucleotides) were synthesized (Tsingke Biotechnology). A standard PCR using KOD-FX DNA polymerase (TOYOBO) was performed to generate transcription templates using the forward primer Spike\_FT (containing T7 promoter sequence) and the reverse primer Spike\_R. A standard transcription was performed using T7 RNA polymerase (New England Biolabs) and the transcripts were cleaned with MicroElute RNA Clean-Up Kit (Omega Bio-Tek). The concentration of the RNA product was quantified using NanoDrop One (Thermo Fisher Scientific). Then, the transcribed RNA was fragmented using a 10 $\times$  fragmentation buffer (New England Biolabs). After fragmentation, the RNA segments were analysed with 2.5% agarose gel. Before detection, the RNA fragments were diluted to 5  $\mu\text{g}/\mu\text{l}$  in 5% saliva of the author.

### Isolation of small RNAs from patient tissues and RT-qPCR

HCC foci and paracancerous tissue samples were obtained during hepatectomy procedures from two HCC patients at Tongji Hospital, yielding four tissue samples in total. The study was conducted with the approval of the Ethics Committee of Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology

(TJ-IRB20230301), and written informed consent was provided by all participants.

The tissues were immediately stored in RNeasy Lysis Solution (Qiagen) and frozen at  $-80^{\circ}\text{C}$ . Small RNA extracts ( $<200$  nt) were prepared using the mirVana<sup>TM</sup> microRNA Isolation Kit (Invitrogen), in accordance with the manufacturer's instructions. A total of 50 mg of each tissue sample was homogenized in 500  $\mu\text{L}$  Lysis/Binding Buffer, using the TissueMaster<sup>TM</sup> High-Throughput Tissue Homogenizer (Beyotime). During this process, a synthetic small RNA (RNA-Ref, 20 nt), amounting to 10 pmol, was spiked into each tissue sample, serving as an external reference to evaluate the efficiency of miRNA recovery and reverse transcription. The final RNA extracts were eluted in 100  $\mu\text{L}$  DEPC water and stored at  $-80^{\circ}\text{C}$  until analysis.

In the RT-qPCR experiments, the first-strand cDNA (20  $\mu\text{L}$ ) was synthesized from 5  $\mu\text{L}$  of purified RNA using the miRcute Plus miRNA First-Strand cDNA Kit (Tiangen Biotechnology). This step involved the use of a poly(A) polymerase to poly-adenylate the target miRNAs and an Oligo(dT)-Universal tag as the reverse transcription primer. The cDNA products were then diluted 1000-fold in DEPC water to be used as qPCR templates. The qPCR analysis was conducted using the miRcute Plus miRNA (qPCR) Kit (Tiangen Biotechnology) on a 7900HT Fast Real-Time PCR System (Applied Biosystems), following the manufacturer's protocols. The RNA-Ref was utilized as the reference to normalize the expression level of miR-122 in both HCC foci and paracancerous tissues. The relative miR-122 expression levels were calculated based on the  $2^{-\Delta\Delta\text{Cq}}$  method (30).

### Generation of unified tandem DNA hairpin probes by synchronized RCA *in situ*

As shown in Figure 1A, the short circular ssDNA templates containing the sequences of targets for RCA were prepared by ligating two stem-loop hairpin units T1 and T2 using T4 DNA ligase (New England Biolabs). The oligo 'T1' was shared in all templates and T2 was sequence-specific (Figure 1A). The ligated products were sequentially treated with Exonuclease III (New England Biolabs) and Exonuclease I (New England Biolabs) to remove all linear DNA strands, leaving only the circular ssDNA templates. All enzymatic reaction products were purified and concentrated using Amicon Ultra-0.5 centrifugal filter devices (Millipore). The obtained ssDNA templates were finally verified by polyacrylamide gel electrophoresis (Supplementary Figure S1) and their concentrations were quantified using NanoDrop One. Each circular ssDNA template contained four functional elements: the docking site exposed in the big loop for primer annealing, the downstream adenine excluded site ('A' excluded site), the adenine core ('A' site) were the critical elements for establishing synchronized amplification procedure, and a sequence-specific region located in the stem was the core element for target recognition (Figure 1A).

The tandem DNA hairpin probes were generated by synchronized RCA *in situ* (Figure 1B). (i) The SH-primers were immobilized to the inner bottom surface of the flow cell. (ii) The circular ssDNA templates were diluted to 20 nM in  $3 \times \text{PBS}$  and annealed to the surface-immobilized SH-

primers in the flow cell at  $37^{\circ}\text{C}$  for 30 min. (iii) The flow cell was rinsed twice with phi29 DNA polymerase reaction buffer to remove all unbound templates. Subsequently, the phi29 DNA polymerase and defective substrate (dATP, dCTP, and dGTP mixture, each at 2 mM) in reaction buffer were added to the flow cell and incubated at  $30^{\circ}\text{C}$  for 5 min. During this period, all working polymerases were paused before the 'A' site (Figure 1A) due to the lacking of dTTP. (iv) The flow cell was rinsed three times again with reaction buffer to remove the free polymerases to allow the subsequent elongation of all probes to start synchronously. After adding the dNTP mix containing dTTP in the reaction buffer to resume the polymerase working, the elongation step was kept at  $30^{\circ}\text{C}$  for 15 min. (v) The reaction buffer containing 100  $\mu\text{M}$  Biotin-11-dUTP (Thermo Fisher Scientific) was added and incubated at  $30^{\circ}\text{C}$  for 1 min. (vi) Finally, the elongation was stopped by 30 mM EDTA. In PBS for 20 min, the single-stranded gap regions of the probe were transferred into double strands by annealing 10 nM 'Flank' ssDNA segments at  $\sim 5$  pN where ssDNA and dsDNA have a similar extension.

All unmodified DNA oligonucleotides were purchased from Tsingke Biotechnology with polyacrylamide gel electrophoresis purification. All modified DNA and RNA oligonucleotides were purchased from Sangon Biotech with high-performance liquid chromatography purification. The sequences of the oligonucleotides were listed in Supplementary Table S1.

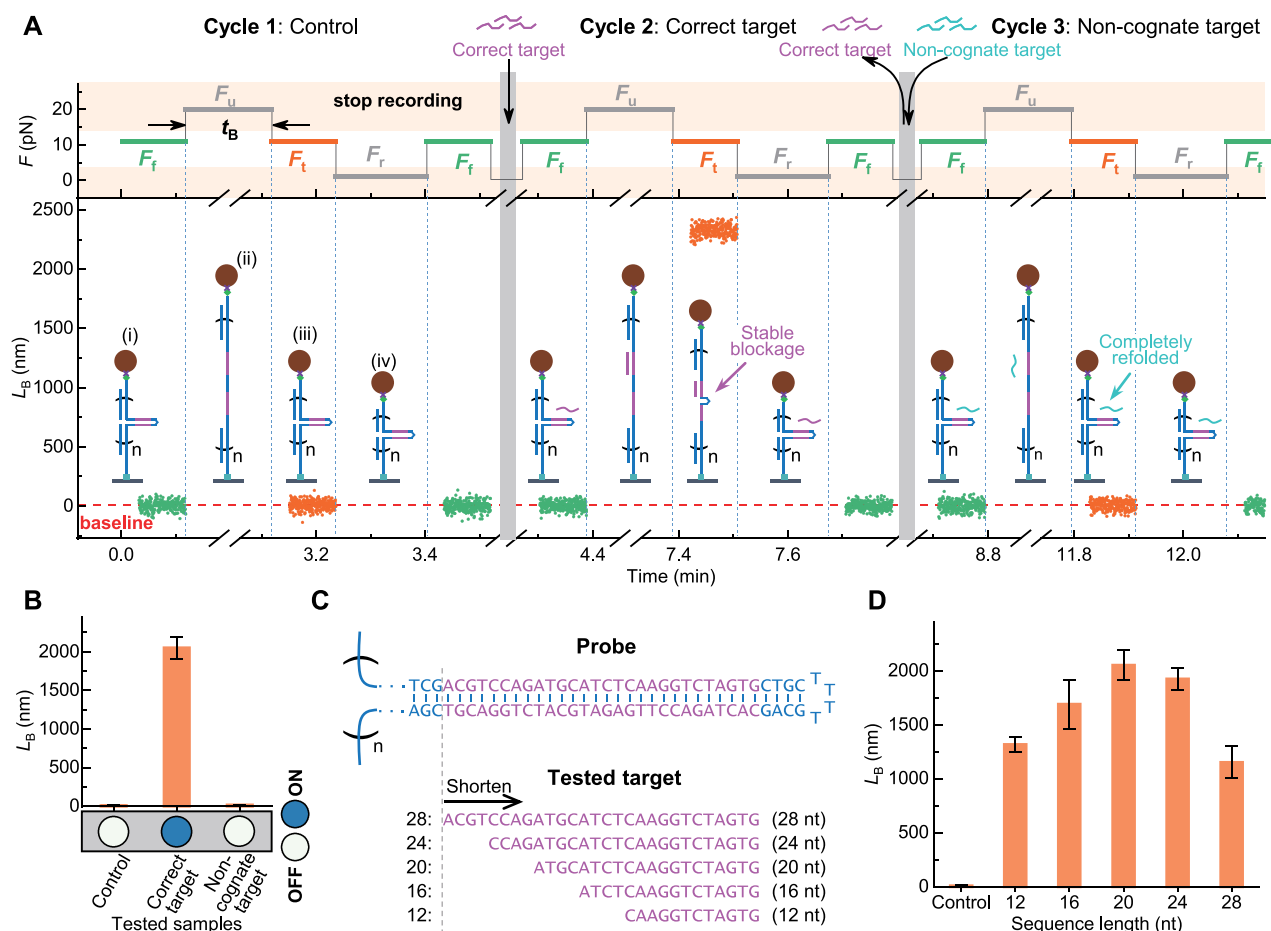
## RESULTS

### Preparation of unified tandem DNA hairpin probes

Due to the limited interaction surface areas of a single probe in single-molecule manipulation experiments, the detection could be effectively achieved at the time scale of about 30 min only when the target concentration exceeds 100 pM (14). To promote sensitivity, we increased the surface area of the interaction surface in the probe by increasing the number of binding sites, which was inspired by previous work (15). We generated unified tandem DNA hairpin probes by synchronized RCA *in situ* (Figure 1). After tethering the probes between the slide surface and the magnetic beads, we monitored the end-to-end distances of the probes in real-time using multiplexed magnetic tweezers (24–26). As shown in Figure 1D, we managed to manipulate multiple probes of similar length in parallel (red squares). We determined that the probes had the polymerization number ( $n$ ) of  $1470 \pm 154$  repeats after 15 min amplification (step iv in Figure 1B) by measuring the total probe extension at 12 pN (Figure 1E and Supplementary Figure S2). The perfectly overlapped force-extension curves (FECs) of different probes further confirmed the unified polymerization number of the probes (Supplementary Figure S3).

### Workflow of THREF reusing the probes

One advantage of THREF is the probes are reusable to detect a series of different samples. We demonstrated the performance of THREF using synthetic single-stranded DNA targets (Figure 2). We repeated force cycles to periodically unfold and refold the hairpins of the tandem DNA probe in



**Figure 2.** Workflow for reusable detections using THREF. The averages and error bars (SDs) of the  $L_B$  were obtained from at least three molecules. (A) Detections of three samples one by one by repeating force cycles. Each force cycle contained four force steps. Cartoon insets showed the probe structures at the four force steps in each force cycle. At the tested force  $F_t$ , a remarkable  $L_B$  was observed only for the correct target. (B) The  $L_B$  in the three detections. The  $L_B$  significantly longer than the control was defined as 'ON'. (C, D) Assessing the blocking efficiencies as a function of the length of targets from 12 to 28 nt. The  $L_B$  of the targets were shown in panel (D).

a solution containing 1 M NaCl at room temperature (Figure 2A). Each force cycle consisted of four consecutive force steps: (i) At the folding force  $F_f$  ( $<13$  pN), the hairpins in the probe existed at the folded state (OFF state) where the sequence-specific regions were concealed in folded hairpins. Here, the extension of the probe was determined to be the baseline (red dashed line in Figure 2A). (ii) After the force increased to the unfolding force  $F_u$  ( $\sim 20$  pN), the probe was fully unfolded and the sequence-specific regions were exposed for target binding. Longer binding time ( $t_B$ ) allowed the binding of more targets. (iii) After the force dropped to the testing force  $F_t$  which equalled  $F_f$ , the extension of the probe relative to the baseline was determined to be  $L_B$ . The hybridization of correct targets (Oligo 20 in Cycle 2) allowed the probe to remain partially unfolded (ON state) due to the stable blockage from hairpin-refolding, which substantially increased  $L_B$ . In contrast, the non-cognate target (Oligo Flank in Cycle 3) could not stably block hairpin-refolding and an extension of the probe restored to the baseline consistent with the control (Cycle 1) was observed. The  $L_B$  from the three testing cycles was plotted as histograms in Figure 2B for clarity. The  $L_B$  is the signal readout of THREF. (iv)

At the end of each cycle, a regenerating force  $F_r$  ( $\sim 1$  pN) was used to dissociate all bound targets and restore the probe to the initial OFF state. At  $F_f$  of the next force cycle, the extension of the probe was restored to the baseline and it was ready for the next test.

Considering the polymerization number of the probe ( $\sim 1500$ ) and the extension of each hairpin monomer (about 70 nm) after fully unfolded, the overall extension of the probe exceeded 100  $\mu\text{m}$ , making the tethered bead too far away from the focus to be imaged (Supplementary Figure S4). Thus, we stopped data recording under  $F_u$ . We also skipped the data under  $F_r$  to further simplify the data (shaded areas in Figure 2A). In addition, we artificially skipped the data in the first two seconds immediately after the force was changed to avoid the hairpin refolding hysteresis caused by any bindings of non-specific pairing or lagging of beads' response (arrowed in Supplementary Figure S5). For illustrative purpose, the actual time-extension trace containing the three force cycles was also plotted in Supplementary Figure S6.

We further demonstrated the stability and reusability of the tandem hairpin probe by repeating  $\sim 2000$  force cycles

lasting for 11 h (Supplementary Figure S7). The excellent stability and reusability of THREF promise rapid and ultimate low-cost nucleic acid detections. In Figure 2A, B, we used 20 nt targets. We next assessed the effect of the length of sequence-specific regions on the detection sensitivities by varying the target length from 12 nt to 28 nt (Figure 2C). We found the sensitivity was improved with the length of the targets up to 20 nt and then decreased at even longer lengths. It is understandable that longer targets hybridize with probes better and increased sensitivity. However, overlong targets (Oligo 28) may form secondary structures which can be predicted by mFold (31) and hinder the hybridization of targets with the probes (Figure 2D and Supplementary Figure S8).

Considering that the Ionic strength, the pH value, and the temperature were known to modulate the stability of the base pairing of nucleic acids, we evaluated the effects of NaCl concentration, MgCl<sub>2</sub> concentration, temperature, and pH on the sensitivity of our method (Supplementary Figure S9). Our findings suggest that higher salt concentrations, elevated temperatures, and a pH of approximately 7.5 enhance the sensitivity of our method. While the further addition of 100 mM MgCl<sub>2</sub> or a higher temperature of 34°C could marginally boost binding efficiency, we opted to use 1 M NaCl, 10 mM Tris-HCl at pH 7.5, and 22°C as the buffer conditions in this study to ensure a longer lifespan for the probes. However, we acknowledge that further optimization of these buffer conditions could be pursued in the future studies.

### THREF achieved femtomolar sensitivity using a single probe without amplification

Disease-associated nucleic acid biomarkers or pathogen-derived nucleic acid fragments in biological fluids have been identified at concentrations ranging from femtomolar to picomolar (32–35), which were well below the LOD of about 100 pM in single-molecule manipulation detection using a single hairpin in 30 min (14). In THREF, we improved the LOD by increasing the number of binding sites in each probe. As shown in Figure 3, we selected the 20 nt targets (Figure 2D) to test the sensitivity of THREF using a single probe. As the concentration of the target increased from 1 fM to 100 pM, we performed sensitivity tests at different  $t_B$  including 0.3, 3 and 30 min (Figure 3A,B) which yielded the LOD of about 321, 57 and 5.5 fM, respectively (Figure 3C). Here, the LOD was the concentration of the target that yielded a  $L_B$  that exceeded the mean background by three times the standard deviation (SD) of the background (36,37). We also achieved a dynamic range over five orders of magnitude by using various  $t_B$  (Figure 3B), which veritably reflected the natural dynamic range of different clinical targets (34). Additionally, the perfect linear relationship between LOD and target  $t_B$  ( $R^2 = 0.986$ ) revealed in Figure 3D suggesting that the THREF had a potential for single-molecule detection after further extending the  $t_B$  (revealed by a dashed extension cord in Figure 3D). At a target concentration of 1 pM and a  $t_B$  of 30 min, we calculated that only about 10% of the sites on the probe were occupied considering the obtained  $L_B$  (~6000 nm) and elastic property of single-stranded

DNA (38–40), which explained the linear relationships between  $L_B$  and  $t_B$  in different target concentrations (Figure 3E), enabling the input target semi-quantification by using THREF.

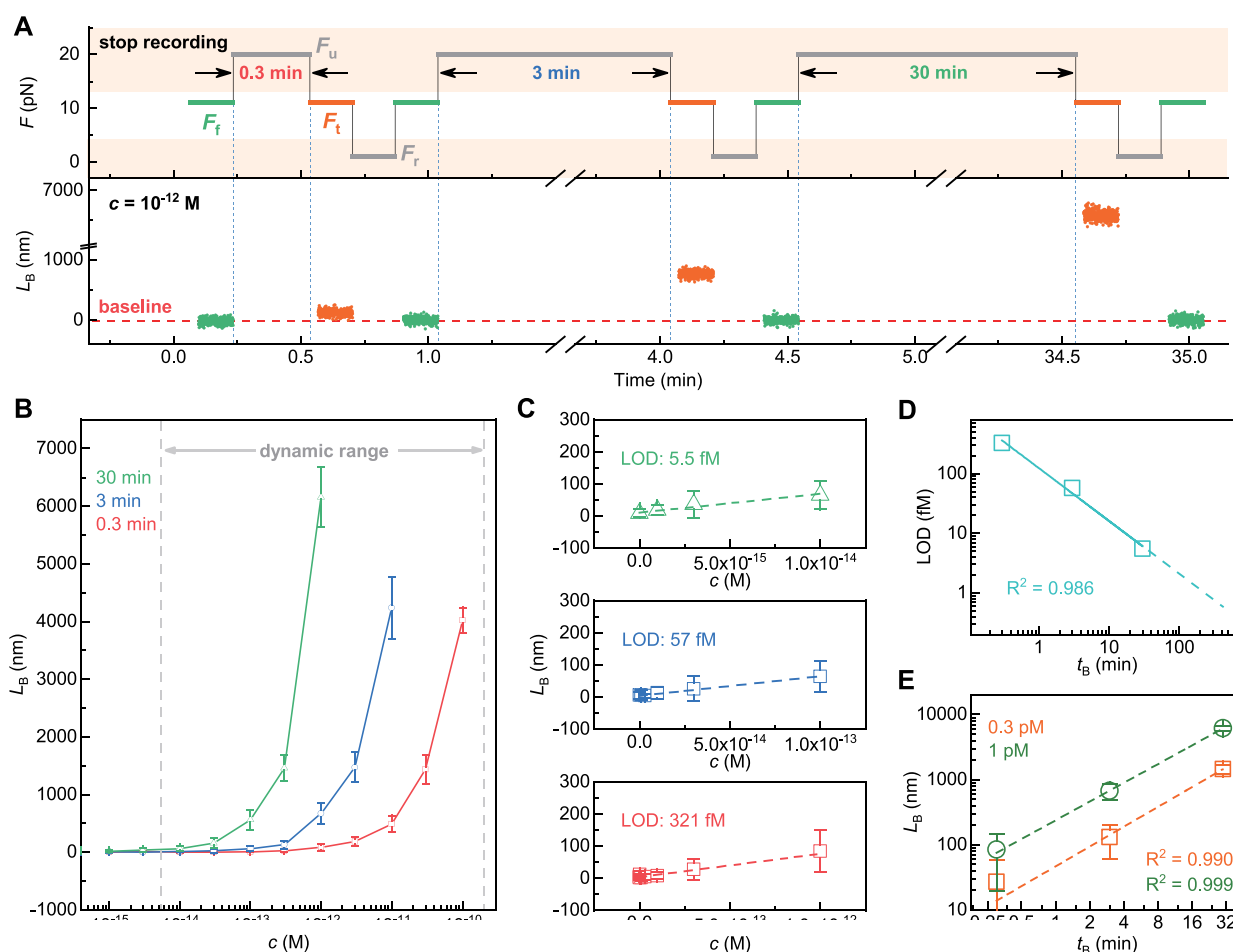
### Detection of miRNA from clinical RNA extract samples.

The miRNAs are endogenous, single-stranded non-coding RNA molecules, typically 18–25 nucleotides in length, that regulate gene expression at the post-transcriptional level. They influence a wide range of biological processes, including development, cell proliferation, and metabolism (41,42). Dysregulation of miRNAs can significantly impact physiology and disease progression (43,44). MiR-122, in particular, is highly expressed in hepatocytes, and its down-regulation has been linked with HCC (45). For this study, we selected miR-122 as a proof-of-principle demonstration, examining the changes in its expression levels in HCC foci tissues using the THREF assay.

To demonstrate the multiplex detection capability of THREF, we parallelly detected miR-122 and the external reference, RNA-Ref, from clinical samples. During the preparation of unified tandem DNA hairpin probes, we mixed the two types of circular ssDNA templates with equal concentrations (10 nM each) before annealing them to the immobilized primers [step (i) in Figure 1B] and then completed the remaining steps of synchronized RCA. Prior to executing the detection, we utilized synthesized testing targets to individually identify each probe. Each testing target effectively inhibited the reziping of its respective probes, thus enabling us to distinguish each probe type. As depicted in Figure 4A, we initially tethered microbeads to two different probe types, p\_RNA-Ref and p\_miR-122. Upon introduction of the first testing target, RNA-Ref, the probes that blocked reziping were identified as p\_RNA-Ref. We then introduced miR-122, and any probes blocking reziping were identified as p\_miR-122.

To evaluate the background signals caused by non-specific interactions between the RNA extract and the probes, we measured the  $L_B$  as a function of the concentration of RNA extract using two types of irrelevant probes (p\_test1 and p\_test2, Supplementary Figure S10). We found the  $L_B$  was negligible when the RNA extract was below 0.2 ng/μL. In the THREF assay, we diluted all RNA extracts 1000-fold to below 0.1 ng/μL, and we performed the experiments in this section using  $F_t = 9$  pN,  $t_B = 3$  min at 1 M NaCl.

Typical time-extension traces for the two parallelly manipulated probe types during the detection were shown in Figure 4B, and the  $L_B$  from six probes were plotted as histograms in Figure 4C. Using RNA-Ref as the reference, we calculated the relative expression of miR-122 in HCC foci and paracancerous tissues from the two patients, respectively (Figure 4D). This calculation involved first obtaining a ratio for each measurement by dividing the  $L_B$  of miR-122 by that of RNA-Ref. Subsequently, we normalized these ratios against the average ratio derived from paracancerous tissues, thus providing us the relative expression of miR-122 (Figure 4D). For both patients, we found miR-122 was down-regulated in HCC foci tissues, which were consistent with the RT-qPCR results (Figure 4E) with the added



**Figure 3.** THREF achieved femtomolar concentration of LOD using a single probe without amplification. The averages and error bars (SDs) of the  $L_B$  were obtained from at least three molecules. (A) Typical time-extension traces using a single probe with 1 pM target (Oligo 20) at the  $t_B$  of 0.3, 3 and 30 min. (B) The  $L_B$  at different target concentrations ranging from 1 fM to 100 pM. (C) Determination of LOD (green for 0.3, blue for 3 and red for 30 min of  $t_B$ ) as the concentration of the target that yielded a  $L_B$  exceeding the background by three times the SD of the background. (D) LOD as a function of the  $t_B$ . (E) The  $L_B$  as a function of the  $t_B$  at 0.3 or 1 pM targets.

advantage of reducing the testing time from longer than 3 h to just 10 min.

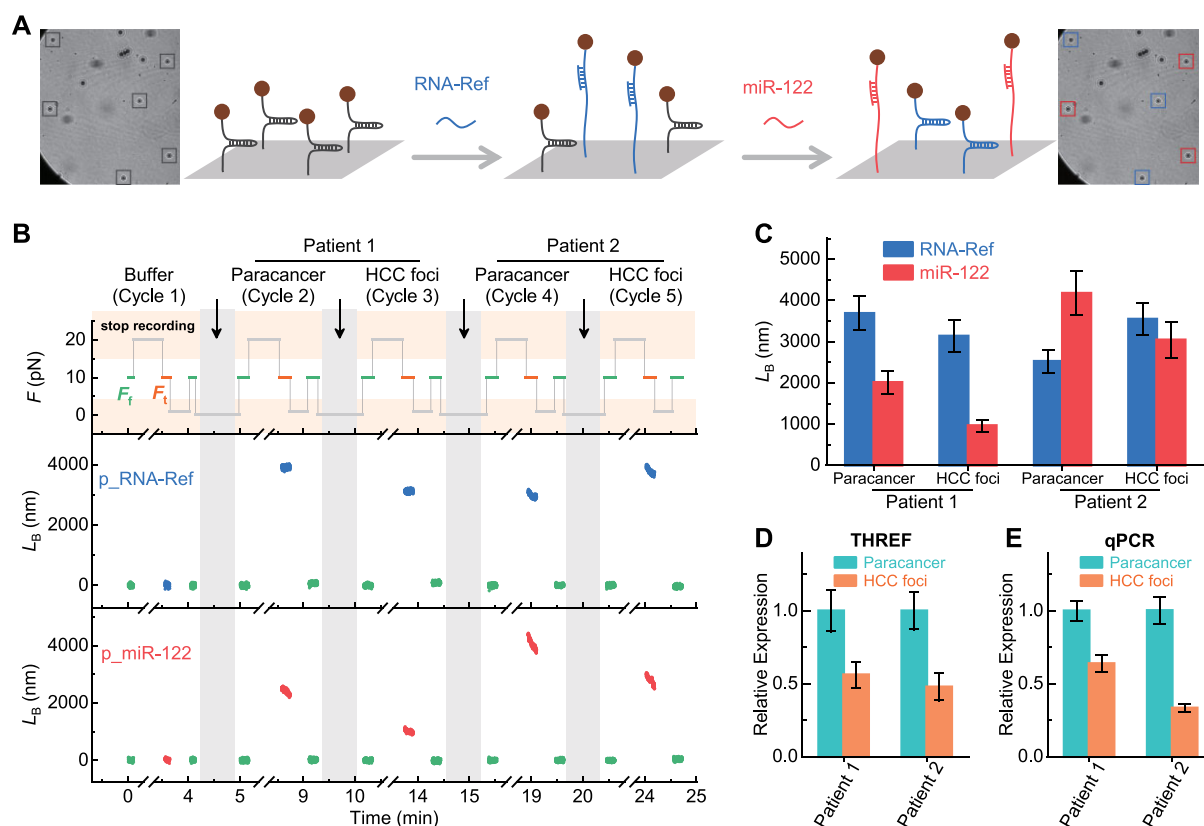
At low concentrations, the  $L_B$  exhibits a linear relationship with the target concentration in the flow cell. After establishing this linear relationship between the  $L_B$  and the concentration of miR-122 (Supplementary Figure S11A), we were able to determine the concentration of miR-122 in the flow cell utilizing the standard curve method (Supplementary Figure S11B).

### Identification of SNVs by THREF

Molecular probes to identify SNVs in DNA or RNA targets are valuable tools for disease diagnosis, therapeutic decisions, and even understanding biological processes (10,46). As shown in Figure 5, we assessed the SNV recognition ability of THREF by detecting the 16 nt correct target (16) or the mutant target with a SNV (v9C). As shown in Figure 5B, at  $F_i$  of 11.5 pN, the  $L_B$  remained stable for the correct target (purple) while gradually shortened for the mutant target with one SNV (olive) due to the unstable blockage. To

further eliminate the  $L_B$  induced by the mutant target, we decreased  $F_i$  from 11.5 to 10.5 and 8.7 pN in the following force cycles (Figure 5C, D). This adjustment of  $F_i$  led to a completely refolded state of the probe for the mutant target whereas maintaining an unfolded state of the probe for the correct target. Thus, we could select an optimized  $F_i$  to distinguish the correct and mutant targets without the need to redesign or reconstruct the probes. As shown in Figure 5, we monitored the same probe and tested different targets sequentially by altering the sample in the flow cell. We analyzed how the length of the sequence-specific region in the probe influences the ability of the THREF method to identify mutant targets in the range of 10–28 nt. Our data indicated that if the SNV position was centrally located in the sequence-specific region of the probe, the THREF method could identify mutant targets when the length of the region was between 14 and 20 nucleotides, with optimal results achieved at approximately 16 nt for this sequence (Supplementary Figure S12).

Inspired by previous studies (2), we further evaluated the ability of THREF to identify mutants by systemat-



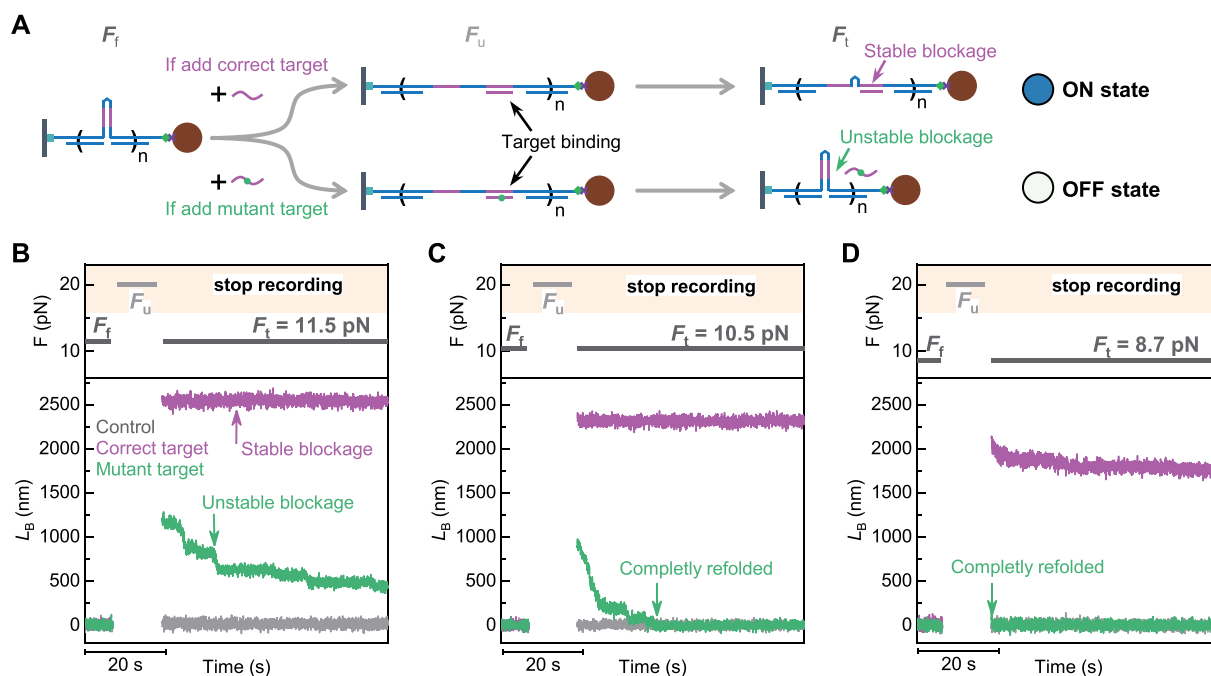
**Figure 4.** Detection of miRNA from clinical RNA extract samples. The averages and error bars (SDs) of the  $L_B$  were obtained from at least three molecules. (A) Pipeline for the identification of probe types before parallel detections. After identification, the probes were marked by coloured squares (blue for p\_RNA-Ref, red for p\_miR-122). (B) Typical time-extension traces for the two parallelly manipulated probe types during the detection. Black arrows indicated the solution change steps in the flow cell. (C) Histograms showing the  $L_B$  for the two probe types (blue for p\_RNA-Ref, red for p\_miR-122) induced by each sample. (D) Relative expressions of miR-122 in HCC foci and paracancerous tissues from the two patients obtained in the THREF assay. (E) Relative expressions of miR-122 in HCC foci and paracancerous tissues from two patients obtained in the RT-qPCR assay.

ically testing the effects of SNV positions, SNV types and SNV numbers (Figure 6, 100 pM targets and  $t_B$  of 10 seconds were used). Three sets of mutant targets were tested. The first set of mutant targets contained different single-nucleotide substitutions at different positions of the target. The second set of mutated targets contained all possible SNV types including substitutions, deletions, and insertions at position nine of the target. The third set contained a different number of substitutions along the SNV-specific region ranging from one to four. To characterize the differences between correct and mutant targets from a thermodynamic perspective, we calculated the free energy differences ( $\Delta G$ ) between probe-hybridization of the correct target and the mutant targets (Figure 6B) using the online tool UNPACK (47). We found most of the mutations gave a positive energy penalty of over 3.1 kcal/mol except the SNVs located at the terminals of SNV-sensitive region which had only one neighbour providing base-stacking based on the nearest-neighbour model (48,49).

To assess the performance of SNV-discrimination of THREF, we measured the  $L_B$  of the probe for all sets of targets individually by scanning  $F_t$  from 2.7 pN to 12.7 pN (Figure 6C–E). We introduced a concept of ‘critical force’ which was defined by the maximum  $F_t$  where  $L_B$  was com-

pletely below a threshold of 30 nm (Figure 6B). Based on the extension of a single unfolded hairpin at  $F_t$ , the  $L_B$  slightly longer than 30 nm indicates that only one binding site of the probe is occupied by the target. We found that this 30 nm threshold effectively excluded any background signals in the absence of target blockage (Supplementary Figure S13). By setting  $F_t$  between the critical force of the correct target and that of the mutant target, the correct target can block the reziping of hairpins in the probe whereas the mutant targets can’t, thus we could distinguish the mutant and correct targets. If we need to distinguish multiple mutant targets from the correct targets,  $F_t$  should be set lower than the smallest critical force among the mutant targets.

By plotting  $\Delta G$  and the critical force together for all mutant targets, we found a strong correlation between them with a coefficient of determination  $R^2 = 0.7$  (Figure 6B, inset). The strong correlation between  $\Delta G$  and the critical force enabled us to design SNV-sensitive sequences of the probe *in silico* by maximizing  $\Delta G$ . To exclude the results of Figure 6 were caused by accidental sequences, we repeated the assessments using another set of target sequences and similar performance of SNV-discrimination was verified (Supplementary Figure S14).



**Figure 5.** Identification of SNVs by THREF. (A) the mechanism of identifying the mutant targets. At appropriate tested forces, hybridization of correct targets stably blocked hairpin-refolding (ON state) whereas hybridization of mutant targets couldn't stably block hairpin-refolding (OFF state) (B–D) Representative time-extension traces at the tested forces of 11.5, 10.5 and 8.7 pN for the correct (purple) and mutant (green) targets.

### Multiplex profiling of let-7 family members in serum

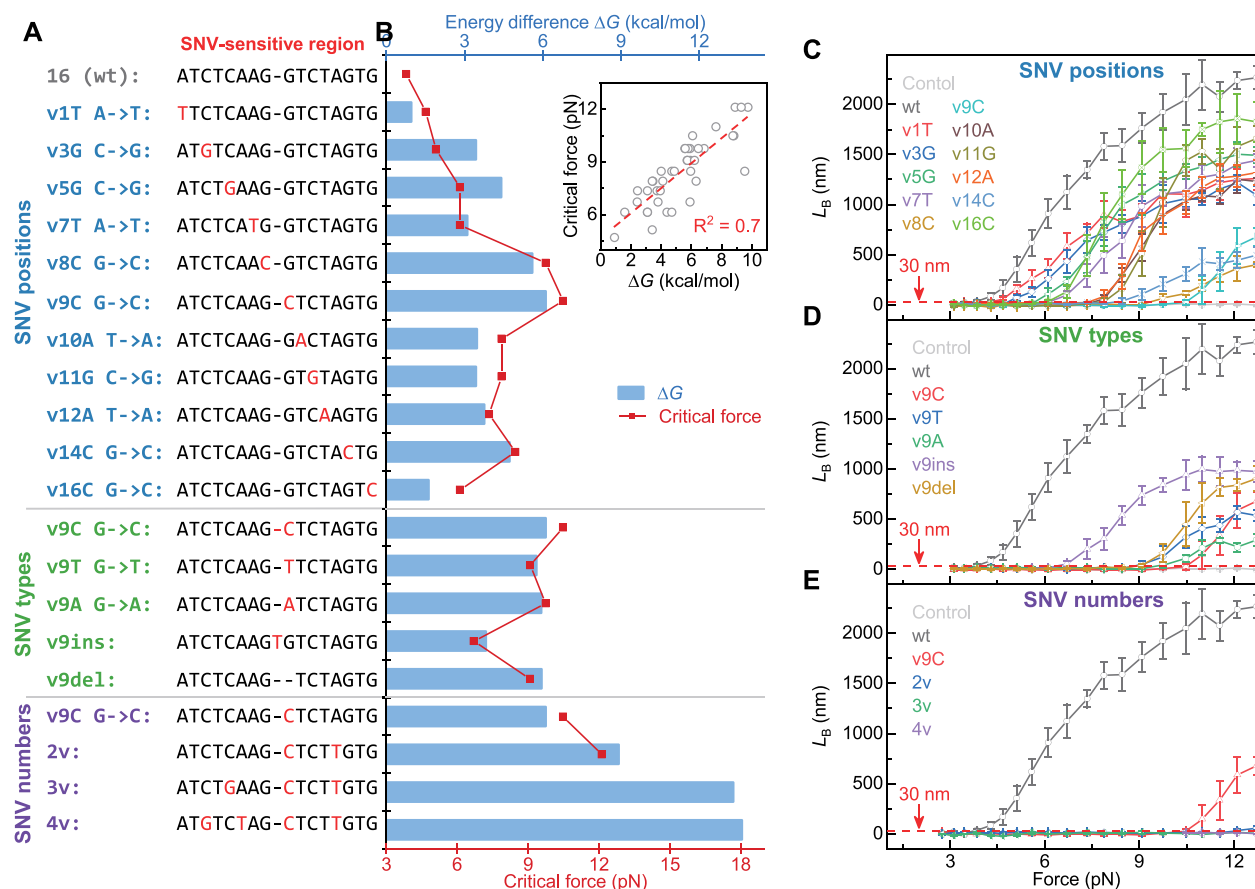
The stable existences of miRNAs in biological fluids such as serum and urine make them non-invasive biomarkers for biological research or disease diagnosis. However, the fast and specific detection of miRNAs was severely hindered by their short lengths and sequence similarities (50,51). The miRNAs of let-7 family were the first discovered human miRNAs with highly conserved sequences (52), differing by only one or two nucleotides among them, such as let-7a, let-7b, let-7c, and let-7d (Figure 7A). Here, to discriminate these four miRNAs using THREF, a set of specific probes named p\_let-7 (p\_let-7a, p\_let-7b, p\_let-7c and p\_let-7d) were generated by integrating the sequences of 16 nucleotides near the 3'-ends, where the mutations are mainly located, into SNV-specific regions, as shown in Figure 7A.

To compare the efficiency of THREF to detect DNA and RNA targets, we measured the  $L_B$  for let-7b miRNA and its DNA analog (D\_let-7b) using the probe p\_let-7b at  $F_t$  in the range of 2.7–12.7 pN. We found the  $L_B$  is slightly short for RNA than DNA but still suitable for detection (Supplementary Figure S15). The weaker detection performance for RNA than DNA targets was consistent with our previous studies that the peeled RNA is difficult to reanneal to the stretched ssDNA to form RNA-DNA hybrids (24,53). As shown in Figure 7B–E (100 pM targets and  $t_B$  of 10 seconds were used), when  $F_t$  was in the range of 5.0–7.0 pN, each probe of p\_let-7 could efficiently discriminate its correct target from the mutant targets indicated by the blocked length longer or shorter than 30 nm (dashed lines), thus enabling the discrimination of these members. The strong false positive signals induced by the target let-7b for p\_let-7c (Fig-

ure 7D) and by the target let-7a for p\_let-7d (Figure 7E) at  $F_t$  larger than 7.0 pN could be explained by the T-G and the G-U wobble base pairs that have comparable thermodynamic stabilities to the canonical A-T Watson-Crick base pair (54,55). The discrimination of wobble base-pairing has always been considered a challenge in previous detection assays (2). Here, we eliminated the false positive signals by adjusting the  $F_t$  to 6.5 pN, further proving the excellent performance in SNV-discrimination of the THREF assay (Figure 7D, E).

To further evaluate the SNV-discrimination performance of THREF assay in complex RNA environments such as when correct and mutant targets co-exist in the solution, we used the probe p\_let-7b to detect 10 pM let-7b mixed with increasing concentrations (0.1–10 000 times of let-7b) of let-7c using the  $t_B$  of 3 min (Figure 7F). We found that the presence of let-7c hardly affected the specific detection of let-7b, even at a fraction of 99.99%. In addition, we also found that the  $L_B$  was remarkably shortened in high concentrations of let-7c (>100 times higher than let-7b). We think that high concentrations of let-7c may hybridize to the probe, occupy the binding sites and reduce the hybridization of the correct miRNA let-7b. When we replaced the let-7c with 100 nM the non-cognate DNA oligo 'Flank', the  $L_B$  for let-7b was not influenced at all (Supplementary Figure S16), which further indicated that the reduction of  $L_B$  with the interference of >100 times of let-7c was caused by occupation the probe binding sites. Nevertheless, we were still capable of discriminating the correct let-7b with the interference of 10 000 times of let-7c or non-cognate oligos.

To discriminate all four types of miRNAs in one reaction in complex biological fluids, we monitored nine probes in



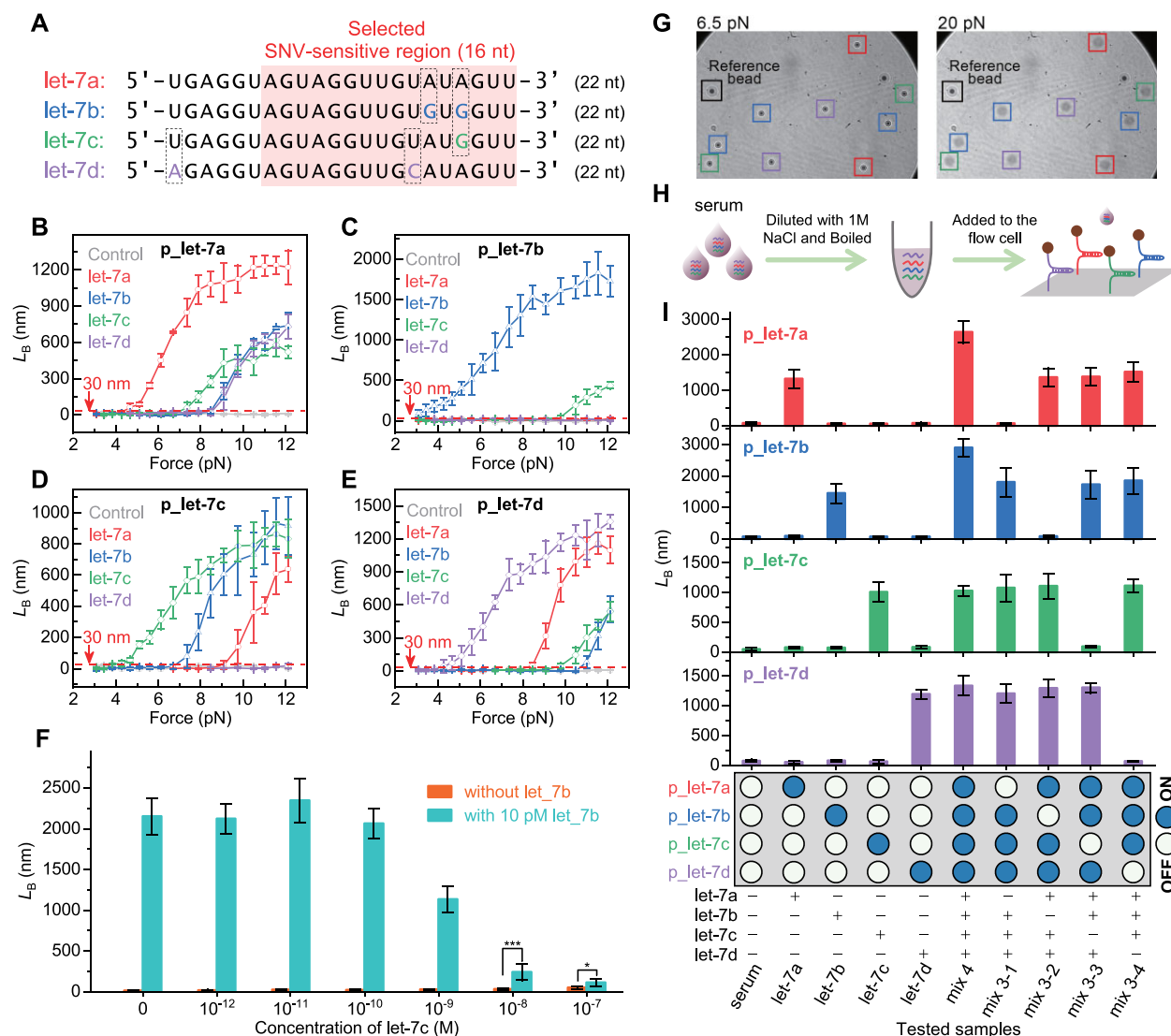
**Figure 6.** The ability of THREF to identify SNVs at various positions, types and numbers of SNVs. The averages and error bars (SDs) of the  $L_B$  were obtained from at least three molecules. (A) The sequences of the correct target and the mutant targets with various positions, types (substitutions, insertions or deletions) and numbers (one to four) of SNVs. (B) The energy differences in probe-hybridization of the targets caused by mutations ( $\Delta G$ ) were plotted as bars. The experimental critical forces for the targets were also plotted as lines and symbols. Inset: the strong correlation between  $\Delta G$  and the experimental critical forces. (C–E) The  $L_B$  induced by each target at a series of tested forces. The red dashed lines at 30 nm represented an empirically determined threshold as described in Supplementary Figure S13.

parallel covering all four types of p-let-7 probes (squared with different colors in Figure 7G). All probes existed at the folded state at 6.5 pN (left panel) while existed at the unfolded state at 20.0 pN (right panel). After we identified the types of probes using testing targets with a similar method shown in Figure 4A, we performed the force cycles to evaluate the discrimination capabilities using the samples containing different combinations of the four miRNAs (Figure 7H–I, 10 pM for each miRNA and 3 min of  $t_B$  were used) in 10% fetal bovine serum (Sangon Biotech). In Figure 7, combining the data obtained from three individual measurements, we found that in all cases, only the correct targets produced significant longer  $B_L$  at the  $F_t$  of 6.5 pN compared to the mutant targets, and switched the ‘OFF’ states of probes to the ‘ON’ states. Note that, we determined whether the target was correct or mutant based on the relative  $B_L$  of the parallel probes rather than the threshold of 30 nm. The representative time-extension traces of the four specific probes in parallel (Supplementary Figure S17) further revealed the multiplexing and reusability of the THREF assay in discriminating close-related miRNAs.

### Genotyping SARS-CoV-2 strains in saliva

All the DNA and RNA targets detected in the above experiments were short in length, optimizing the hybridization efficiency by avoiding the influence of DNA/RNA secondary structures (Figure 2D) (56). To further expand the application scenarios of THREF, by discriminating the RNA fragments in saliva, we genotyped three SARS-CoV-2 strains: wild-type (Wuhan-Hu-1), B.1.617.2 (Delta), and BA.1 (Omicron). We discriminated five target sequences located in two different loci of SARS-CoV-2 *spike* gene: Spike 614 site (A23403G, S:D614G, occurred both in B.1.617.2 and BA.1), and Spike 679/681 site (T23599G, S:N679K and C23604A, S:P681H, occurred in BA.1; C23604G, S:P681R, occurred in B.1.617.2). For comparison, the wild-type strain lacks all the above mutations. Sequence differences between the five targets for the three strains were aligned in Figure 8A.

After integrating the selected SNV-specific regions (Figure 8A) into five types of probes (named starting with ‘p’ in Figure 8F and Supplementary Figure S18) by synchronized RCA, we tested their performance in SNV-discrimination using the DNA analog targets (Supplementary Figure S18)

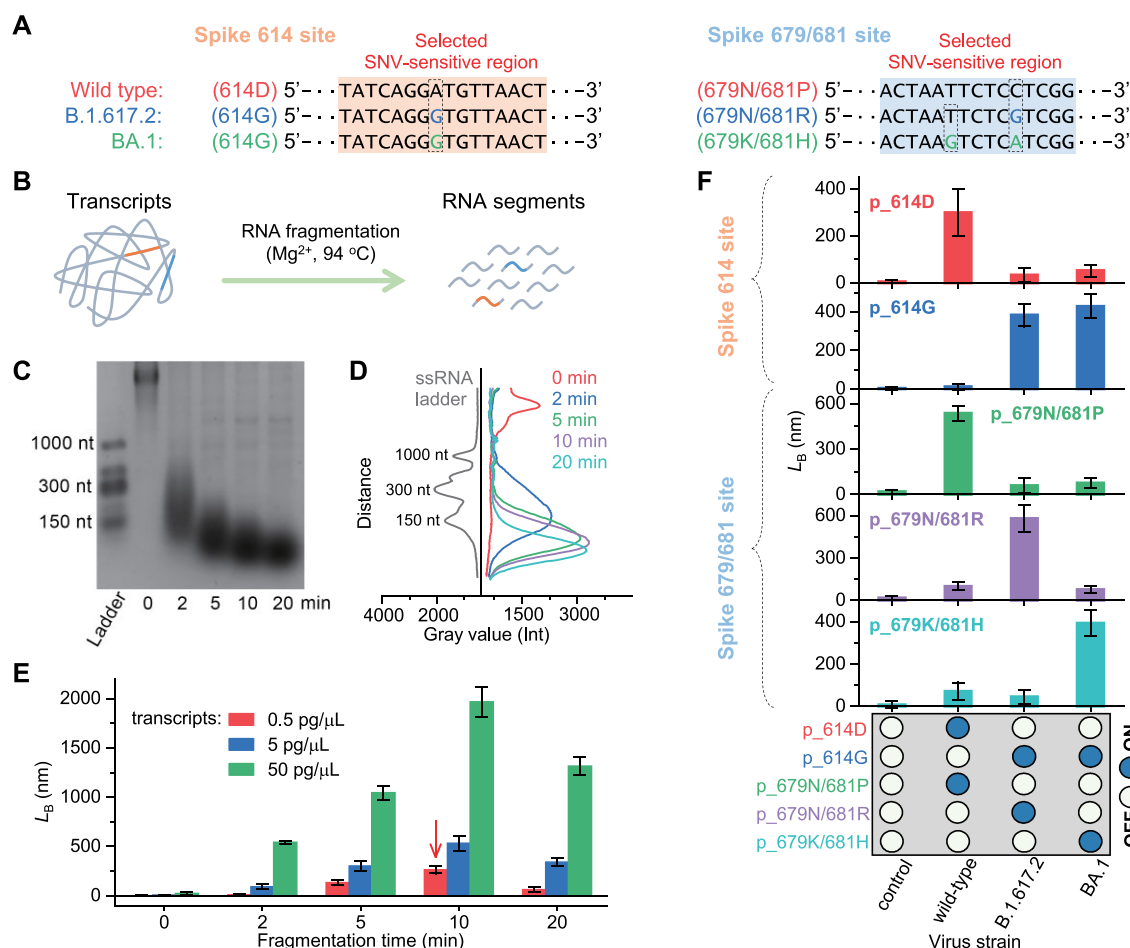


**Figure 7.** Multiplex profiling of let-7 family members in serum. The averages and error bars (SDs) of the  $L_B$  were obtained from at least three molecules. (A) Aligned sequences of the four let-7 family miRNAs. The locations of SNVs were marked by rectangles. The 16-nt sequences used for the design of probes were marked by a red background. (B–E) Manipulating the four types of probes (p\_let-7a, p\_let-7b, p\_let-7c and p\_let-7d) in parallel to detect the four miRNAs in turn. The  $L_B$  were plotted as functions of tested forces. The red dashed lines were the thresholds of 30 nm. (F) The  $L_B$  using the probe p\_let-7b to detect let-7b mixed with increasing concentrations of let-7c. Continued t-test P-values indicated the statistical significance (\* $P < 0.05$ , \*\*\* $P < 0.001$ ). (G) A field-of-view showing parallel manipulation of the four types of probes: p\_let-7a (red), p\_let-7b (blue), p\_let-7c (green), and p\_let-7d (purple). All probes existed at the folded state at 6.5 pN (left panel) and unfolded state at 20.0 pN (right panel) indicated by the diffraction patterns of the magnetic beads. At 20 pN, the magnetic beads are far away from the focal plane and thus their diffraction patterns dispersed thoroughly. (H) Pipeline for the amplification-free and purification-free identification of miRNAs in serum. (I) The discrimination capabilities using the samples containing different combinations of the four miRNAs in 10% FBS. The  $L_B$  were normalized as an ON/OFF map for visualization.

and optimized the  $F_t$  to 6 pN for largely eliminated all the false positive signals. Moving beyond the synthetic short DNA analog targets, we sought to establish a detection procedure against long RNA transcripts to further mock clinical samples which were expected to have strong secondary structures that may interfere with our detection. Inspired by previous work (56), we introduced an RNA chemical fragmentation strategy and randomly fragmented the long RNA into small pieces at lengths depending on the fragmentation time in the range of 2–20 min (Figure 8B–D). In the THREF assay, we found the fragmentation time of 10 min had the best performance (Figure 8E, the  $t_B$  of

30 min was used). At the optimal fragmentation time of 10 min, we observed an apparent  $L_B$  at the concentration of 0.5 pg/ $\mu$ L RNA fragments (red arrow in Figure 8E).

We next detected the fragmented RNA (5 pg/ $\mu$ L) from three SARS-CoV-2 strains in 5% human saliva by monitoring all five types of probes in parallel (Figure 8F). We found that all three strains induced at least five times longer  $L_B$  for the strain-specific probes (indicated as ‘ON’ state) than the  $L_B$  for the non-specific probes (indicated as ‘OFF’ state). Note that we determined the strain types based on comparing the intensity of  $L_B$  produced by strain-specific probes and non-specific probes. One advantage of the



**Figure 8.** Genotyping SARS-CoV-2 strains in saliva. The averages and error bars (SDs) of the  $L_B$  were obtained from at least three molecules. (A) Aligned sequences of the two selected SNV-sensitive regions of three SARS-CoV-2 strains. SNVs in the sequences were marked by rectangles. (B) Schematic of the fragmentation of viral RNA transcript. (C, D) Fragmentation analysis of wild-type RNA transcript that was fragmented at 94 °C for different reaction times. (E) The  $L_B$  as functions of the fragmentation time detected using the p\_614D probe. (F) Genotyping SARS-CoV-2 strains in 5% saliva at 5 pg/μL RNA using the  $t_B$  of 30 min. The corresponding  $L_B$  were normalized as an ON/OFF map for visualization.

parallel THREEF assay is that we could eliminate the slight off-target signals by comparing the signal intensities generated by the specific probes and the non-specific probes.

## DISCUSSION

In this work, we developed the THREEF assay without amplification or fluorescence for nucleic acid detection with SNV-discrimination by combining synchronized RCA and multiplexed magnetic tweezers. We detected targets at femtomolar concentrations within 30 min with a single probe. We managed to discriminate SNVs including the hard-to-detect G–U or T–G wobble mutations (2). Using mock clinic samples, we profiled the let-7 family miRNAs in serum and genotyped SARS-CoV-2 strains in saliva.

To improve the sensitivity, instead of amplification of the nucleic acids in the sample in most previous assays, we generated unified tandem hairpin probes each containing ~1500 binding sites by synchronized RCA *in situ* (Figure 1E). The immobilized synthesis of the probe on the glass surface was shown to be more structurally stable

than post-amplification immobilization, as discussed previously (29). The spatial independence of the extending molecules during the immobilized synthesis avoided their mutual interferences due to thermal fluctuations and prevented nascent single-stranded DNA from forming entangled winding structures due to undesirable pairings, which greatly improved the availability of the products, especially for those with large-size (more than 200 kb) and lots of palindromic sequences. We found that THREEF using a single probe had a sensitivity of femtomolar using 30 min  $t_B$  (Figure 3C) which outperformed many amplification-free detection methods including Northern blotting, Microarray, and DNA Nanodevices (57–59) and achieved the concentrations of clinical or biological samples ranging from femtomolar to picomolar (32–35). The linear relationship between LOD and  $t_B$  ( $R^2 = 0.986$ , Figure 3D) suggested that THREEF had the potential to improve the sensitivity by further extending the  $t_B$ . In addition, the sensitivity may also be improved by monitoring multiple probes parallel to increase the number of binding sites. If twenty probes with similar polymerization numbers were monitored in

parallel and each probe contained  $\sim 1500$  binding sites, we could monitor 30000 binding sites simultaneously.

The probes with similar lengths allow us to use multiplexed magnetic tweezers to manipulate multiple probes in parallel (Figures 1D, 4, 7G–I, 8F, and Supplementary Figure S17), thereby realizing multi-target co-detection: one was to parallelly quantify the concentrations of miR122 from clinic HCC tissues (Figure 4), one was to simultaneously detect multiple members with closely related sequences of the let-7 miRNA family in mixed samples (Figure 7) and the other was to simultaneously detect not only various regions of the viral genome but also multiple mutation types in each region, that finally enabling rapid genotyping different SARS-CoV-2 strains in a single reaction (Figure 8). By manipulating probes for different viruses, the THREF was also expected to well address the clinical needs of multi-virus co-detection, especially respiratory viruses with similar clinical symptoms. In this work, we detected only five types of targets simultaneously (Figure 8F) due to the field-of-view was limited by the  $100\times$  oil objective but could be easily expanded by about 7-fold if using of  $40\times$  oil objective. The precision to determine the position of the bead in  $z$  direction (i.e. the extension of the probe) was  $\sim 2$  nm when  $100\times$  objective was used and was  $\sim 6$  nm when  $40\times$  objective (NA1.3) was used (60). The precision of  $\sim 6$  nm is enough for the THREF assay.

Hairpins or other structures usually be designed in the nucleic acid probes to make the binding of both correct and mutant targets less energetically favourable (2,49,61). This allows for the establishment of a sensitive equilibrium where slight energy differences caused by SNVs can result in substantial differences in hybridization specificities, enabling SNV discrimination. However, this necessitates intricate probe design and iterative adjustments of detection conditions (2,49), causing excessive consumption of public resources and hindering multiplex detections. In contrast, our assay relied on a novel mechanism that introduced an additional regulation factor: mechanical energy. This allowed us to flexibly compensate for energy deficiencies even in the case of highly robust probe designs. In our work, we designed a much longer stem structure (55–57 bp) compared to the sequence-specific region (10–28 base pairs) into the hairpin (Figure 1A), making the folded state (the hairpins in the probe are completely folded) significantly more stable than the blocked state (refolding of the hairpin is blocked by the binding of the target) in the absence of external force (Supplementary Figure S19A). Upon application of external force ( $F_t$ ), this mechanical energy tilts the energy landscape, compensating for the energy difference between the folded and the blocked states (Supplementary Figure S19B). Considering that the SNV in the mutant target incurs an essential energy penalty ( $\Delta G$ ) during probe-hybridization and reduced the blocking ability, we can optimize  $F_t$  for each probe in pre-experiments to tilt the energy landscape and ensure that the folded state was more stable than the blocked state for the correct target but less stable for the mutant target, allowing successful differentiation between correct and mutant targets (Supplementary Figure S19B).

As a proof-of-concept, many robustly designed probes in our study were tested with targets containing SVNs and

obtained larger critical forces ranging from 6 pN to over 12.5 pN compared to correct targets ( $< 4$  pN) except for the two end-mutated ones in Figure 6, enabling us to better distinguish the correct targets from mutations without any cross-talk signals, even the hard-to-detect G–U or T–G wobble mutations (Figure 7C, D) (2). In addition, the robust and ultra-specific detection assay implemented by introducing a large excess of mutant targets into solutions further promised the application of our THREF assay in rare mutant allele detections with an estimated specificity of more than 99.99% (Figure 7F), which was enabling the detection of one mutant target in a background of ten thousand wild-type targets (19,20,62).

Furthermore, we also demonstrated the robustness of reaction conditions by spiking synthesized miRNAs and SARS-CoV-2-derived RNA fragments into FBS and human saliva, respectively. All the cases showed comparable detection performance compared to pure salt solutions (Figures 7I, 8F). In general, the diagnostic assays will benefit from solution robustness because clinical samples or other enzymatic products can be detected directly and rapidly without any purification steps or buffer-exchange procedures (Figure 7H). By combining with the stability and reusability of the tethered probe (Supplementary Figure S7), we even verified the possibility of minimizing the material cost of the detection.

While the rapid detection feasibility of the THREF method has been demonstrated using clinical samples (Figure 4), there are certain limitations to be addressed. First, the THREF requires magnetic tweezers and expertise in data analysis, which might limit its deployment in point-of-care or resource-limited scenarios. Automation of the magnetic tweezers and standardization of the data analysis workflow can somewhat mitigate this limitation. Second, contaminants in clinic samples may lead to false-positive results due to non-specific interactions. In this work, the re-folding pressure of the DNA hairpin effectively eliminates bindings of SNV targets with lower stabilities (Figure 7F), thereby endowing the THREF method with high specificity. However, high concentration of background RNA in RNA extracts can still introduce non-specific binding signals and reduce the detection sensitivity (Supplementary Figure S10). Furthermore, contaminants such as proteins bound to the hairpins may prevent probe recycling, an issue that could potentially be solved by a regeneration buffer containing the protein denaturant sodium dodecyl sulfate (SDS) (63). Third, for the detections of low-abundance targets, a target amplification procedure may be required before the detections. In the case of THREF, the targets need to be amplified into ssDNA by asymmetric PCR or into ssRNA by transcription. As a proof-of-concept, we combined THREF with asymmetric PCR and determined the genetic polymorphism of CYP3A5, which has applications in personalized tacrolimus dosing guidance (64). We determined the CYP3A5 genetic polymorphism in the blood samples of three patients by identifying the nucleotide at the SNV position (6986A  $>$  G; rs776746) (Supplementary Figure S20). Our THREF results revealed that patient 1 had A/A, patient 2 had A/G, and patient 3 had G/G nucleotides at the SNV positions of the allele, respectively (Supplementary Figure S20C–E). These findings were in accordance

with the clinical Sanger sequencing outcomes (Supplementary Figure S20F). Besides amplification, target enrichment is also an effective method to enhance the sensitivity of detecting rare targets. For instance, Wang et al. developed a CRISPR-based method for target DNA enrichment (17). In addition, high-throughput single-molecule manipulation platforms that manipulate thousands of probes simultaneously could also increase the sensitivity (23,65–68).

## DATA AVAILABILITY

The data underlying this article are available in the article and its online supplementary material.

## SUPPLEMENTARY DATA

Supplementary Data are available at NAR Online.

## FUNDING

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